Frank Brase

Senior Software Architect | Embedded C++, Distributed Systems & GenAI

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PROFILE

Hands-on software architect with 20+ years of experience in Modern C++, embedded Linux (Yocto) and distributed systems. Combines architectural responsibility with active software development, CI/CD automation and technical knowledge sharing. Uses AI-assisted development tools for rapid prototyping and technical evaluation.

PROFESSIONAL EXPERIENCE

Software Architect (Hands-on)

ITK Engineering, Holzkirchen | 12/2022 – 07/2026

- Design and hands-on implementation of software architectures and UI-intensive embedded systems for high-end consumer and industrial products
- Definition and implementation of system concepts: SDX, microservices, cloud-native architectures, containers, Zephyr RTOS
- GenAI-assisted rapid prototyping using GitHub Copilot, Cursor and Claude Code
- Technical enablement through architecture training, internal talks and presales support in connectivity and telemetry

Embedded Software Developer

ITK Engineering, Munich | 02/2014 – 12/2022

- Hands-on development in C++, C, Python, Rust, Go, Java and C# across customer projects
- Project domains: industry, automotive, medical robotics and aerospace
- Visualization and telemetry of vehicle and device data on embedded Linux platforms
- UI development using Qt/QML, GTK4, Flutter and React
- Member of a dedicated trainer group teaching junior developers the C++ software engineering workflow, including pull requests and code reviews

Software Developer

Alcatel-Lucent (now Nokia), Nuremberg | 07/2000 – 01/2014

- Software development for mobile base stations in high-performance environments
- Integration of emerging network stacks for first-call solutions in international expert teams

CORE SKILLS

Architecture: Embedded & Distributed Systems · Microservices · Cloud-native · arc42 · SDx
Development: Modern C++ · C · Python · Java / Spring Boot · Rust · Go · C# · Embedded Linux / Yocto · VxWorks · Zephyr RTOS · Bare-metal platforms
UI & HMI Architectures: Qt/QML · Flutter · GTK4 · React · Angular · Vue
Engineering & DevOps: GitHub Actions (CI/CD) · Git · Docker · Static Analysis · Code Coverage
Generative AI: GitHub Copilot · Cursor · Claude Code · LangChain · LangGraph · RAG · Azure OpenAI · LangSmith

EDUCATION & CERTIFICATIONS

Degrees: Diploma in Mathematics (Computer Science), Univ. Hannover · MSc Applied Mathematics, Brunel
Certifications: iSAQB-F Software Architect · ISO 26262 · ISO 13485 · Generative AI with GitHub Copilot
Languages: German (native) · English (fluent) · Spanish (advanced)